llmXive Automated Review of Identifying stimulus-driven neural activity patterns in multi-patient intracranial recordings

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The manuscript provides a comprehensive survey of methods for identifying stimulus-driven neural activity. Most claims are well-supported, but a few specific factual assertions regarding numerical statistics and methodological definitions require verification against the cited sources to ensure strict accuracy.

In Section 1.2.1, the text states the cerebral cortex comprises "roughly 80% of the adult human brain's mass, but only roughly 20% of its neurons," citing Herculano-Houzel (2009). While this paper is the seminal work on total neuron counts (~86 billion) and isometric scaling, the specific 80/20 mass-to-neuron ratio is a derived statistic not explicitly highlighted as a primary finding. The authors should verify if this specific percentage is directly stated in the source or if it requires a qualifier.

Similarly, in the same section, the text claims the brain contains "roughly 100 billion glial cells," citing Azevedo et al. (2009). This paper famously refutes the old "10:1" glia-to-neuron ratio, proposing instead a near 1:1 ratio (approx. 86 billion of each). Citing this work to support a "100 billion" figure for glia creates a numerical inconsistency with the source's primary conclusion. The authors should clarify the specific counts supported by these references.

Additionally, Section 1.2.1 claims that "Stimulus-driven responses in individual neurons... can only be measured using invasive approaches like iEEG and ECoG." This is an overstatement. While iEEG and ECoG are invasive, true single-unit recordings in humans often utilize microelectrode arrays (e.g., Utah arrays) or depth electrodes with micro-wires, which are distinct from standard clinical iEEG/ECoG grids. The text conflates specific modalities with the broader category of invasive single-unit recording.

Finally, in Section 2.1.2, the description of Representational Similarity Analysis (RSA) specifies computing the ”element-wise correlation” (implying Pearson) between matrix upper triangles. While common, the foundational paper by Kriegeskorte et al. (2008) explicitly discusses Spearman rank correlation and Kendall’s tau as robust alternatives. Presenting Pearson as the singular method without acknowledging the flexibility emphasized in the source text is a minor over-simplification.

Addressing these points will ensure the manuscript’s claims align precisely with the evidence provided in the citations.

- [**writing**] Section 1.2.1: The claim that the cortex is 80% mass but 20% neurons cites Herc09. Verify if this specific ratio is explicitly in that paper or if it is a derived statistic requiring clarification.
- [**writing**] Section 1.2.1: The text cites AzevEtal09 for ’100 billion glial cells,’ but that paper argues for a ~1:1 ratio (~86B each). The ’100B’ figure contradicts the cited source’s main finding. Clarify the numbers.
- [**writing**] Section 1.2.1: Claiming single-neuron responses ’can only be measured using... iEEG and ECoG’ is inaccurate. Microelectrode arrays (e.g., Utah arrays) also measure single units invasively but are distinct from standard iEEG/ECoG.
- [**writing**] Section 2.1.2: The RSA description implies Pearson correlation is the standard method. However, KrieEtal08b emphasizes Spearman/Kendall as robust alternatives. Acknowledge this flexibility to align with the source.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
Figure 1 provides a useful conceptual overview of resolution trade-offs but lacks a visual legend for the color-coded methods and uses a misleading linear scale for data spanning many orders of magnitude.

Figure 2
Figure 2 effectively visualizes simulated neural signals, but the caption is truncated and fails to describe Panels C and D. Additionally, the complete absence of axis scales or units across all panels limits the figure’s scientific utility.

Figure 3
Figure 3 is a clear, well-organized schematic that effectively illustrates the conceptual distinctions between univariate/multivariate, units/networks, and within-brain/across-brain analyses. The visual layout is intuitive, and the caption provides sufficient context to interpret the diagrams without ambiguity.

Figure 4

Figure 4 effectively illustrates the concept of discrete stimulus timeseries. Panel A clearly shows the sequence of stimuli (faces, scenes, objects) interspersed with blank intervals, while Panels B and C accurately depict the corresponding binary timing and category-specific identity signals. The visual elements align perfectly with the provided caption.

Figure 5

Figure 5 illustrates continuous stimulus timeseries but lacks critical axis labels and a defined color scale in Panels B and C, undermining the interpretation of feature values and trajectories described in the caption.

Figure 6

The figure effectively visualizes the density difference between single-patient and multi-patient electrode coverage, but it fails to include a color legend for the 53 patients in Panel B and lacks explicit row labels (A/B) to match the caption structure.

Figure 7

The figure effectively visualizes the geometric trajectories described in the caption, but the caption text is truncated, and Panel C lacks a descriptive explanation in the legend.

Figure 8

Figure 8 is severely compromised by a truncated caption that cuts off the explanation for Panel B and a raw LaTeX artifact ('$$s') in the text. Additionally, Panel A lacks a colorbar, rendering the quantitative values of the heatmap uninterpretable.

Figure 9

The rendered image and the provided caption for Figure 9 are completely mismatched; the image shows a pipeline of electrode locations and correlation matrices, while the caption describes a Gaussian process regression model. Additionally, the heatmaps in the image lack necessary legends or colorbars to interpret the data.

- [**writing**] Figure 1: The caption states 'Green shading denotes in vitro methods' and 'Blue shading denotes invasive in vivo methods,' but the figure itself contains no legend or key to map these colors to the specific methods (e.g., Patch clamp, iEEG, ECoG) shown. The reader must infer the color coding solely from the caption text.

- [**science**] Figure 1: The caption explicitly notes 'axes are not drawn to scale,' yet the figure uses a linear grid layout for both spatial and temporal axes. This is misleading because the data spans orders of magnitude (e.g., nanoseconds to decades, angstroms to decimeters); a linear representation distorts the relative resolution differences between methods like Patch clamp and fMRI.

- [**writing**] Figure 2: The caption text for Panel B is incomplete, ending abruptly with 'recording surface' and missing the description for Panels C and D.

- [**writing**] Figure 2: Panels C and D display oscillatory waveforms but lack any caption description or definition of what these signals represent.

- [**science**] Figure 2: The axes in all panels lack tick marks, numerical scales, or units, making it impossible to verify the 'arbitrary units' claim or the temporal resolution.

- [**science**] Figure 5: Panel A's x-axis is labeled 'Time' with ticks 0 and 99, but the caption describes a continuous stimulus without specifying the time unit or scale, making it impossible to assess autocorrelation claims.
- [**science**] Figure 5: Panels B and C show 3D trajectories and heatmaps but lack axis labels for the 3D plot axes and no colorbar/legend mapping grayscale intensity to feature values, despite the caption discussing 'values'.
- [**writing**] Figure 5: The grayscale colorbar in Panel C is present but unlabeled; the caption does not define what the grayscale range (0 to 1) represents, leaving the scale ambiguous.
- [**science**] Figure 6: Panel B displays a dense cloud of multi-colored dots representing 5023 electrodes from 53 patients, but the figure lacks a color legend or key to map specific colors to individual patients, rendering the 'Colors denote different patients' claim unverifiable.
- [**writing**] Figure 6: The caption states 'A. Example patient' and 'B. Across-patient...', but the rendered image lacks explicit 'A' and 'B' labels to distinguish the two rows of panels.
- [**writing**] Figure 7: The caption text is truncated at the end ('C. Interpreting coord'), cutting off the description for Panel C.
- [**science**] Figure 7: Panel C displays images labeled 'Original' and 'Reconstruction' but lacks a corresponding legend entry in the caption to explain what these images represent or how they relate to the geometric models in Panels A and B.
- [**fatal**] Figure 8: The caption text is truncated mid-sentence at the end ('The per-image weig'), preventing the reader from understanding the description of Panel B.
- [**science**] Figure 8: Panel A contains a raw LaTeX formatting artifact ('$$s') in the caption instead of the intended symbol (likely 'x' or 'x_s'), making the reference to factor centers illegible.
- [**science**] Figure 8: Panel A displays a color heatmap but lacks a colorbar or scale, making it impossible to interpret the magnitude of the 'spherical factors' or 'radial basis functions' shown.
- [**fatal**] Figure 9: The rendered image displays panels labeled A through E (Electrode locations, RBF interpolation, Correlation matrices, Merged model, Observed activity), but the provided caption describes a completely different figure (Building across-patient models using Gaussian process regression) with panels A and B only. The visual content and text are mismatched.
- [**science**] Figure 9: The rendered image lacks a colorbar or legend to interpret the 'Location' heatmaps in Panels B, C, and D, making the intensity values and correlation strengths illegible.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript exhibits a moderate level of jargon density, particularly in the sections detailing specific analytical methods (Sections 41.1.1 and 41.2.1). While the text is generally accessible to a neuroscience audience, several acronyms and specialized terms are introduced without sufficient definition for a broader, non-specialist readership.

Specifically, in Section 41.1.1, the terms ”Generalized Linear Models” and ”Multivariate Pattern Analysis” are introduced, but the subsequent heavy reliance on the acronyms ”GLMs” and ”MVPA” without a clear, explicit definition at the very first instance (e.g., ”Generalized Linear Models (GLMs)”) creates a barrier for readers unfamiliar with these specific abbreviations. Similarly, in Section 41.2.1, ”Inter-subject correlation” and ”Inter-subject functional correlation” are introduced, but the acronyms ”ISC” and ”ISFC” are used immediately after without a clear, standalone definition sentence that a non-expert could easily reference.

The term ”procrustean transformation” in Section 41.2.1 is a significant hurdle. While ”Procrustes analysis” is a standard term in statistics and geometry, the phrasing ”procrustean transformation” is less common and may be opaque to readers outside of specific mathematical or statistical sub-fields. A brief explanatory phrase (e.g., ”a geometric alignment method known as Procrustes analysis”) would significantly improve accessibility.

Additionally, the term ”timeseries” is used frequently. While common in signal processing, standard English usage prefers ”time series” (two words). The inconsistent usage or the potential perception of ”timeseries” as a jargonized form should be addressed to maintain clarity for a general scientific audience.

Finally, ”LFP” (Local Field Potentials) is defined in Section 41.1.2, but the text later uses ”LFP” and ”LFPs” in a way that assumes the reader has retained the definition from several paragraphs prior. Re-stating the full term or ensuring the acronym is clearly anchored at the first use in each major subsection would be beneficial.

Overall, the paper is well-written for its target audience of neuroscientists, but a few targeted edits to define acronyms and clarify specialized terms would make it more inclusive for a broader scientific readership.

- [**writing**] Define ’GLM’ (Generalized Linear Models) and ’MVPA’ (Multivariate Pattern Analysis) at their first occurrence in Section 41.1.1. While the full terms are used initially, the acronyms appear frequently thereafter; ensure the first instance explicitly states ’Generalized Linear Models (GLMs)’ and ’Multivariate Pattern Analysis (MVPA)’ to aid non-specialist readers.”

- [**writing**] Replace the term ’procrustean transformation’ in Section 41.2.1 with ’Procrustes analysis’ or provide a brief parenthetical explanation (e.g., ’a geometric alignment method’) upon first use. The current phrasing assumes familiarity with a specific statistical term that may exclude readers from adjacent fields.”

- [**writing**] In Section 41.1.2, define ’LFP’ (Local Field Potentials) immediately after the first mention. The text introduces ’Local field potentials (LFPs)’ but later uses ’LFPs’ and ’LFP’ interchangeably without re-establishing the acronym for readers who may have skipped the initial definition.”

- [**writing**] In Section 41.2.1, clarify ’ISC’ and ’ISFC’ (Inter-subject correlation and Inter-subject functional correlation) at first use. The text introduces the full terms but relies heavily on the acronyms in subsequent sentences; ensure the expansion is clear and the acronym is explicitly defined for the first time."

- [**writing**] In Section 41.1.1, replace 'timeseries' with 'time series' (two words) for consistency with standard English usage, or define it as a specific technical term if the single-word form is intended as a distinct jargon. The current usage varies and may confuse non-specialists."

paper_reviewer_logical_consistency

Verdict: minor_revision

The manuscript presents a comprehensive survey of methods for identifying stimulus-driven neural activity, but there are specific logical gaps where the conclusions do not strictly follow from the stated premises or where definitions are introduced but not integrated into the subsequent argumentation.

First, in Section 2.1 ("How can we measure neural 'activity'..."), the text explicitly defines "brain activity" to include non-neuronal phenomena, specifically citing glial cells and their role in metabolism and synapse formation. The text then states, "the term neural activity can be more precise way of referring to these phenomena." However, the subsequent sections reviewing specific methodologies (GLMs, MVPA, RSA, etc.) focus exclusively on neuronal signals (spikes, LFPs, BOLD). The logical inconsistency arises because the review implies these methods are sufficient to capture the "brain activity" defined in the introduction, yet the methods described are incapable of measuring the glial activity just defined as part of that scope. The conclusion that these methods identify "stimulus-driven neural activity patterns" is sound only if the definition of "brain activity" is narrowed to "neural activity" immediately, or if the text explicitly acknowledges that the reviewed methods ignore the glial component of the previously defined "brain activity."

Second, in Section 3.1.3 regarding "Joint stimulus-activity models," the authors argue that neither the stimulus trajectory nor the neural trajectory alone provides a complete reflection of internal mental representations. The text posits that the "true representation" lies "somewhere between" the stimulus and neural trajectories, leading to the conclusion that a joint model is the "most accurate" approach. This conclusion is not logically necessitated by the premises. The premises establish that the stimulus model is limited by experimentalist assumptions and the neural model is limited by noise and internal state. However, it does not follow that the truth is a linear or geometric intermediate (the "between" state). It is equally plausible that the neural trajectory is a complex, non-linear function of the stimulus trajectory, or that the stimulus model is entirely irrelevant to the specific neural representation being measured. The text asserts the necessity of a joint model without providing a mechanism or logical derivation for why the "truth" must be a compromise between the two rather than a specific transformation of one. This weakens the logical support for the claim that joint models are inherently superior to refined univariate models.

Finally, in Section 3.2.1 regarding "Hierarchical matrix factorization models," the text claims that HTFA allows for "across-subject comparisons" because "each subject has the same set of factors." The logical leap here is that having the same *number* of factors (K) and a shared *prior* (global template) guarantees that the factors are *comparable* in a functional sense. The text assumes that the hierarchical constraint forces the factors to align functionally across subjects. However, without a premise stating that the underlying neural architecture is invariant across subjects (which is contradicted by the earlier discussion of individual differences in electrode placement and neuroanatomy), the conclusion that the resulting factors are directly comparable is not fully supported. The method enforces a structural similarity, but the logical link to functional comparability requires an additional assumption about the stability of functional topography across the patient population, which is not explicitly stated as a premise.

- [**writing**] Section 2.1 defines 'brain activity' to include glial metabolism but later reviews methods (GLMs, RSA) that only measure neuronal signals. Clarify if the methods are insufficient for the defined scope or if the definition should be narrowed to 'neural activity' to maintain logical consistency.
- [**science**] Section 3.1.3 claims joint models are 'most accurate' because truth lies 'between' stimulus and neural trajectories. This assumes a linear intermediate without proving why the true representation isn't a non-linear transform of one or the other. Provide a mechanism for this assumption.

paper_reviewer_overreach

Verdict: minor_revision

The manuscript is a comprehensive survey chapter rather than an empirical study presenting new data. Consequently, the primary risk of overreach lies in the rhetorical framing of the text, which occasionally blurs the line between reviewing existing literature and claiming novel insights or definitive superiority of specific methods.

First, the Abstract and Introduction (lines 24-45) list a wide array of methods (GLMs, MVPA, RSA, etc.) and state, "we will consider a variety of... approaches." While the intent is to review, the phrasing "Examples from the recent literature serve to illustrate..." is slightly ambiguous. It risks implying the chapter contains original illustrative examples or a novel comparative framework. Given the text is a review, the language should be tightened to explicitly state that the chapter *synthesizes* existing examples rather than *providing* them as new evidence.

Second, in Section 3.2.2 (Representational Similarity Analysis), the text claims RSA "can sometimes be a more sensitive way of identifying stimulus-driven neural patterns (e.g., compared with GLMs and MVPA)." This is a strong comparative claim regarding statistical power and sensitivity. While likely true in specific noisy contexts, the text presents this as a general property of the method without immediately qualifying it with the specific conditions (e.g., high noise, non-linear mappings) or citing the specific study that demonstrated this superiority in iEEG. This risks overgeneralizing a context-dependent finding.

Third, the concluding remarks (Section 5, lines 1080-1082) state: "As a field, cognitive neuroscience is still decades away from being able to link neural and stimulus features at high levels of detail." This is a speculative temporal prediction. While the author's expertise lends weight to this, the chapter provides no data or trend analysis to justify the specific "decades" timeline. This extrapolates beyond the scope of a methodological review. The statement should be reframed as a current significant challenge or a long-term goal, rather than a fixed prediction of the field's trajectory.

Finally, the text occasionally uses definitive language for methods that are still evolving. For instance, in Section 3.2.3, the description of Joint Stimulus-Activity Models implies a settled geometric framework ("The resulting aligned coordinates may then be directly compared"). While the math is sound, the practical application in multi-patient iEEG is fraught with alignment issues (as noted in Section 2.3). The text should ensure it does not overstate the ease or universality of these geometric solutions without reiterating the limitations of electrode coverage and anatomical variability discussed earlier.

Overall, the paper is a solid review, but the authors should carefully audit the text to ensure that claims of methodological superiority, future timelines, and the nature of the "examples" provided are strictly framed as reviews of existing work rather than new empirical findings.

- [**writing**] The abstract and introduction list numerous specific modeling approaches (GLMs, MVPA, RSA, etc.) as if the chapter presents novel applications or comparative results. However, the text is a methodological review. The language should be adjusted to clearly frame these as 'reviewed approaches' rather than implying the chapter itself performs these analyses on new data.
- [**writing**] In Section 3.2.2, the text states that RSA 'can sometimes be a more sensitive way' than GLMs/MVPA. This is a strong comparative claim. The text should explicitly cite the specific literature demonstrating this sensitivity advantage in the context of noisy iEEG data, rather than presenting it as a general, unqualified fact.
- [**writing**] The conclusion states the field is 'decades away' from linking neural and stimulus features at high levels of detail. While a reasonable opinion, this specific temporal prediction ('decades') is an extrapolation not supported by the data or analysis presented in the chapter. This should be softened to reflect it as a current challenge rather than a fixed timeline.

paper_reviewer_safety_ethics

Verdict: accept
The manuscript is a methodological survey chapter rather than an empirical study reporting new human data collection. Consequently, it does not present primary IRB/IACUC concerns, as the authors are synthesizing existing literature and describing established analytical frameworks (e.g., GLMs, RSA, Hyperalignment) rather than detailing a specific new experimental protocol involving human subjects.

The text appropriately contextualizes the ethical and clinical constraints of intracranial EEG (iEEG) research. Specifically, the "Invasive approaches" section (lines 230–255) correctly identifies that such recordings are restricted to neurosurgical patients (typically with drug-resistant epilepsy) and emphasizes that electrode placement is driven by clinical necessity, not research goals. The manuscript explicitly notes the limitations this imposes on generalizability and the inherent risks of the procedure, satisfying the requirement to acknowledge the vulnerable nature of the study population.

Regarding data privacy and consent, the paper discusses the use of "multi-patient" data and "across-participant" models (Section 4.2) but does not present raw patient data or specific datasets that would require a new consent review. The references to external datasets (e.g., citing \cite{EzzyEtal17}, \cite{SedeEtal03}) imply that the original studies adhered to necessary ethical standards. The manuscript does not propose any dual-use applications (e.g., neural decoding for non-consensual surveillance or mind-reading) that would raise safety alarms; the described technologies (decoding speech, identifying stimulus categories) are standard in cognitive neuroscience and are framed within the context of understanding brain function and potential clinical applications like brain-computer interfaces.

No conflicts of interest are apparent in the text provided. The discussion of "Human-in-the-loop" techniques and automated modeling (Section 3.2) is methodological and does not suggest the deployment of these systems in high-stakes, unregulated environments. The review concludes that the manuscript adheres to standard ethical norms for a review chapter in this field.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The manuscript functions as a methodological survey chapter rather than an empirical research article presenting new data. Therefore, the standard metrics for scientific evidence (sample size calculations, p-values, effect sizes, replication rates) are not directly applicable to the text itself, as no new experiments are conducted. The "evidence" provided consists of a synthesis of existing literature.

However, from the perspective of evidence strength, the manuscript lacks a systematic review protocol. The selection of methods discussed (GLMs, MVPA, RSA, HTFA, etc.) and the specific citations used to illustrate them appear to be based on the author's expertise rather than a reproducible search strategy with defined inclusion/exclusion criteria. This limits the ability to assess whether the review is comprehensive or subject to selection bias. For instance, in Section 4.2, the text asserts the benefits of Hierarchical Topographic Factor Analysis (HTFA) and Gaussian Process models for multi-patient data, citing specific papers (e.g., Owen et al., 2020; Kumar et al., 2021). While these citations are valid, the review does not summarize the sample sizes (number of patients/electrodes) or the statistical robustness of the findings in those source papers. Without this context, the reader cannot evaluate the strength of the evidence supporting the claim that these methods are "ideally suited" or superior to alternatives for the specific problem of variable electrode coverage.

Furthermore, the discussion of "challenges" in Section 3.1 regarding electrode coverage (Figure 1) relies on a specific dataset (Ezzyat et al., 2017) to illustrate the sparsity of coverage. While illustrative, the review does not quantify the variability in coverage across the broader literature or discuss how sample size (number of patients) impacts the reliability of the "stitched" full-brain maps mentioned. The evidence presented is conceptual and illustrative rather than quantitative. To strengthen the scientific evidence base of this survey, the authors should either provide a brief summary of the sample sizes and statistical power of the key studies cited for each method or explicitly state that the review is a narrative summary of expert opinion rather than a systematic evidence synthesis.

- [**science**] The manuscript is a methodological survey and does not present primary empirical data, sample sizes, or statistical tests. Consequently, standard evidence metrics (power, effect sizes, p-values) are not applicable to the text itself. However, the review of external literature lacks a systematic search strategy or inclusion criteria, making the evidence base for the claimed ”comprehensive” overview difficult to verify or replicate.
- [**science**] Several methodological claims regarding the superiority or specific utility of techniques like HTFA or Gaussian Process models (e.g., Section 4.2) rely on citations to external studies (e.g., Owen et al., 2020) without summarizing the sample sizes, patient heterogeneity, or statistical robustness of those foundational studies. The review should briefly contextualize the evidence strength of the cited works to support the recommendations.

paper_reviewer_statistical_analysis

Verdict: minor_revision
This manuscript serves as a methodological survey rather than an empirical study reporting new statistical results. Consequently, there are no specific p-values, confidence intervals, or effect sizes to audit directly. However, the statistical rigor of the *described* methodologies requires clarification to ensure readers can apply these techniques correctly without inflating Type I error rates.

The primary concern is the absence of a discussion on **multiple-comparisons correction**. The text details high-dimensional analyses, such as searchlight RSA (Section 41.3.1.3) and whole-brain ISC (Section 41.3.2.2), which inherently involve testing thousands of hypotheses simultaneously. Without explicit mention of False Discovery Rate (FDR) or Family-Wise Error Rate (FWER) control methods (e.g., Bonferroni, cluster-based permutation tests), the described workflows risk generating spurious ”significant” findings. The authors should add a paragraph in the relevant sections or a general ”Statistical Considerations” subsection outlining standard correction practices for these specific modalities.

Secondly, the **significance testing procedures** for correlation-based metrics are under-specified. For RSA (Section 41.3.1.3) and ISC/ISFC (Section 41.3.2.2), the text explains how correlation matrices are computed but does not describe how statistical significance is determined. In naturalistic stimulus paradigms, standard parametric assumptions (e.g., independence of timepoints) are often violated due to temporal autocorrelation. The authors should specify the recommended null models (e.g., phase-randomized surrogates, trial-shuffling) and permutation strategies required to generate valid p-values and confidence intervals for these statistics.

Finally, regarding the **Gaussian Process (GP) models** (Section 41.3.2.3), the text describes the interpolation of activity from sparse electrodes to a full-brain grid. It is critical to address **model validation and overfitting**. The authors should clarify how kernel hyperparameters (e.g., length scales, noise variance) are optimized (e.g., via marginal likelihood maximization) and whether cross-validation is employed to ensure the reconstructed ”SuperEEG” signals generalize beyond the training data. Without this, the reliability of the reconstructed full-brain maps remains theoretically unverified.

Addressing these points will ensure the survey provides a statistically sound roadmap for researchers applying these complex methods to intracranial data.

- [**science**] The manuscript describes statistical methods (GLMs, RSA, ISC) but lacks a dedicated section on multiple-comparisons correction. Given the high dimensionality of intracranial data (thousands of electrodes/timepoints) and the use of searchlight analyses, the authors must explicitly state how false discovery rates (FDR) or family-wise error rates (FWER) are controlled to prevent spurious findings.
- [**science**] In the description of Representational Similarity Analysis (RSA) and Inter-Subject Correlation (ISC), the text mentions computing correlations between matrices or time series but omits details on significance testing. The authors should specify the permutation strategies or null models used to generate p-values and confidence intervals for these correlation coefficients.
- [**science**] The section on Gaussian Process regression (SuperEEG) describes the kernel and interpolation but does not address model validation. The authors should clarify how hyperparameters (e.g., length scales) are selected and whether cross-validation or out-of-sample testing was used to prevent overfitting when reconstructing full-brain activity from sparse electrodes.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript presents a comprehensive and well-structured overview of methods for identifying stimulus-driven neural activity in intracranial recordings. The logical flow from defining neural activity and stimulus models to discussing specific analytical approaches is clear and effective. The writing style is generally academic and accessible, successfully guiding the reader through complex methodological concepts.

However, the text contains several minor but distracting typographical and grammatical errors that should be corrected before publication. Specifically, in Section 2.1, the word ”absense” is used instead of ”absence” when describing feature vectors. In Section 3.1.1, the phrase ”a the M-dimensional” contains a redundant article. Section 1.4 suffers from a repetition error (”participant participant”), and the Summary section misspells ”drug-resistant” as ”drug-resistent.” Additionally, the verb ”proceed” is misspelled as ”procede” in Section 2.1.

While these errors do not fundamentally obscure the scientific meaning, their presence reduces the overall polish and professionalism of the manuscript. A careful proofreading pass to catch these specific instances of typos and grammatical slips is recommended. The rest of the prose is coherent, and the transitions between sections are smooth.

- [**writing**] Correct the typo ’absense’ to ’absence’ in Section 2.1 (Building explicit stimulus models), paragraph 3, where the text describes multivariate real-valued feature vectors.

- [**writing**] Fix the grammatical error 'a the M-dimensional' to 'the M-dimensional' in Section 3.1.1 (Generalized linear models), where the GLM equation variables are defined.
- [**writing**] Remove the redundant word 'participant' in Section 1.4 (Static versus dynamic measures), where the text reads 'across repeated presentations to a single participant participant'.
- [**writing**] Correct the typo 'drug-resistent' to 'drug-resistant' in the Summary section (Section 5), where the text discusses the limitations of the subject population.
- [**writing**] Fix the typo 'procede' to 'proceed' in Section 2.1, paragraph 4, where the text discusses making informed decisions based on prior work.